Analysis of Organic Contaminants in New Zealand Marine Mammals

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Table 1 Biometric data for the individual marine mammals analysed in this study.

Lab#	Doc#	Species	Sex	Age (y)	Location
MM8-HD2	9162	Hector's	Female	1	Waikouaiti
MM8-DD1		Dusky			Kaikoura
MM8-DD2		Dusky			Kaikoura
MM8-DD4		Dusky			Kaikoura
MM9-HD1	8730	Hector's	Female	3	Pegasus Bay
MM8-HD1	9363	Hector's	Female	1	North of Christchurch
MM9-HD2	8736	Hectors'	Female	10	Pegasus Bay
MM9-HD3	8737	Hector's	Female	19	Pegasus Bay
MM9-HD4	8739	Hector's	Male	18	South of Timaru
MM10-HD1	8844	Hector's	Female	2	Pegasus Bay
MM10-HD2	8845	Hector's	Male	1	Akaroa
MM10-HD3	8846	Hector's	Male	1	North of Timaru
MM11-HD1	W004	Hector's	Male	1	North of Westport
MM11-HD2	W005	Hector's	Male	1	Westport
MM11-HD3	W007	Hector's	Male	1	Granity

Table 2 Concentration of individual PCB congeners (ng/g) in New Zealand marine mammals. Congeners are numbered according to the IUPAC convention.

Congener	MM8-HD2	MM8-DD1	MM8-DD2	MM8-DD4	MM9-HD1	MM8-HD1	MM9-HD2	MM9-HD3	MM9-HD4	MM10-HD1	MM10-HD2	MM10-HD3	MM11-HD1	MM11-HD2	MM11-HD3
PCB#77[a]	66	52	14	3.3	47	64	35	61	190	30	40	70	110	170	70
PCB#126[a]	1091	53	65	5.1	1032	667	121	1000	740	400	600	800	130	150	250
PCB#169[a]	393	37	109	14	95	95	39	170	140	100	110	160	50	70	90
PCB#28	5.46	2.58	2.6	0.96	27.3	12.1	3.11	15.2	9.2	9.09	7.43	9.93	6.4	9.1	3.98
PCB#52	13.0	34.0	20.4	5.38	19.8	13.0	1.41	15.6	16.6	8.41	11.6	17.1	4.34	4.64	3.39
PCB#101	58.0	83.2	43.1	6.93	58.3	30.2	4.92	41.6	30.6	27.1	25.1	34.7	7.07	8.12	7.54
PCB#99	89.9	93.1	26.1	9.13	43.3	42.3	3.86	43.8	88.3	16.7	25.9	92.1	14.6	5.99	12.1
PCB#118	439	237	108	19.4	195	182	18.1	216	419	98.0	142	304	82.7	31.8	62.0
PCB#105	122	49.1	31.8	6.39	51.8	48.0	5.5	57.1	127	25.9	51.9	119	23.5	8.91	19.4
PCB#153	1495	1854	567	161	721	666	121	892	1587	300	199	n.d.[b]	170	87.1	133
PCB#138	1204	1154	427	124	519	458	86.2	592	1139	227	201	632	163	74.1	128
PCB#187	285	424	108	34.6	109	93.9	31.6	129	270	47.5	39.9	290	37.5	14.2	37.6
PCB#183	103	84.7	36.3	12.1	42.3	34.5	11.4	52.8	105	17.1	13.7	118	11.9	5.29	11.5
PCB#180	435	430	165	54.0	154	125	45.5	200	390	71.0	61.8	385	55.5	23.2	54.5
PCB#170	157	164	60.7	18.2	71.4	61.7	20.2	94.4	158	31	26.6	181	19.7	7.59	15.7
PCB#202	16.9	24.7	5.48	2.7	6.68	5.08	2.34	8.26	14.2	3.21	1.73	23.1	2.36	0.98	2.36
PCB#194	68.3	71.9	22.0	8.1	19.8	18.4	9.53	33.7	58.1	10.1	5.45	56.3	5.06	2.39	3.52
SUM (μg/g)	4.49	4.71	1.62	0.46	2.04	1.79	0.37	2.39	4.41	0.89	0.81	2.26	0.60	0.28	0.49
TE (pg/g)	175	40.2	23.9	3.94	131	92.5	15.5	120	136	54.4	81.4	130	24.9	20.2	34.8
Potency (μg TE/g PCB)	39	8.5	15	9	64	52	42	50	31	61	100	58	42	72	71

[a] Concentrations for PCB#77 PCB#126 and PCB#169 in pg/g. Concentrations for all other congeners in ng/g.
[b] n.d. = Not Determined due to mass spectrometer technical difficulties.

Table 3. Concentrations (pg/g) of individual PCDD and PCDF congeners in New Zealand marine mammals.

Congener	MM8-HD2	MM8-DD1	MM8-DD2	MM8-DD4	MM9-HD1	MM8-HD1	MM9-HD2	MM9-HD3	MM9-HD4	MM10-HD1	MM10-HD3
2,3,7,8-TeF	2.1	< 0.1	0.35	< 0.4	1.6	2.02	1.8	0.7	1.59	2.08	1.86
non-2,3,7,8-TeF	5.66	1.44	0.35	< 0.3	< 0.8	3.36	< 0.3	< 0.8	< 0.6	< 0.4	< 0.4
2,3,7,8-TeD	4.8	< 0.6	< 0.3	< 0.2	7.02	3.18	< 0.6	5.23	3.98	3.51	3.85
non-2,3,7,8-TeD	< 0.2	< 0.3	< 0.2	< 0.4	< 2	< 0.2	< 0.7	< 1	< 2	< 1	< 1
1,2,3,7,8-PeF	0.67	< 0.3	< 0.2	< 0.1	< 1	< 0.2	< 0.5	< 0.9	< 0.8	< 0.6	< 0.5
2,3,4,7,8-PeF	28.1	< 0.2	< 0.2	< 0.1	12.6	5.84	2.2	6.7	18.9	5.81	7.17
non-2,3,7,8-PeF	7.5	2.5	0.67	< 0.2	1.43	0.92	< 0.7	2.55	8.32	< 0.6	4.86
1,2,3,7,8-PeD	11.4	< 0.2	< 0.3	< 0.2	8.47	3.91	0.94	4.92	5.73	4.72	4.01
non-2,3,7,8-PeD	< 0.1	< 0.3	< 0.1	< 0.2	< 0.8	< 0.1	< 0.6	< 0.5	< 0.5	< 0.4	< 0.3
1,2,3,4,7,8-HxF	< 0.2	< 0.3	< 0.2	< 0.2	< 0.5	< 0.2	< 0.4	< 1	< 0.8	< 0.5	< 0.4
1,2,3,6,7,8-HxF	0.29	< 0.3	< 0.2	< 0.2	< 0.5	< 0.3	< 0.4	< 1	0.81	< 0.6	< 0.5
2,3,4,6,7,8-HxF	0.57	< 0.4	< 0.4	< 0.3	< 0.3	0.28	< 0.3	< 0.9	< 0.9	< 0.5	< 0.6
1,2,3,7,8,9-HxF	< 0.2	< 0.2	< 0.1	< 0.2	< 0.7	< 0.2	< 0.6	< 0.9	< 1	< 0.7	< 0.3
non-2,3,7,8-HxF	4.2	3.38	2.89	< 0.2	< 1	1.17	1.21	4.17	6.77	< 0.9	2.46
1,2,3,4,7,8-HxD	0.80	< 0.3	< 0.2	< 0.2	< 1	< 0.4	< 0.5	< 0.9	< 0.9	< 0.7	< 0.6
1,2,3,6,7,8-HxD	1.98	< 0.3	< 0.5	< 0.2	3.22	1.12	0.77	2.32	< 1	< 2.0	< 0.7
1,2,3,7,8,9-HxD	< 0.2	< 0.2	< 0.1	< 0.2	< 1	< 0.4	< 0.6	< 2	< 1	< 0.8	< 0.7
non-2,3,7,8-HxD	< 0.6	< 0.5	< 0.4	< 0.4	< 3	< 0.5	< 1	2.33	2.73	< 0.7	< 1
1,2,3,4,6,7,8-HpF	0.23	0.50	0.43	< 0.1	1.66	< 0.2	0.63	1.79	1.5	< 0.8	< 1
1,2,3,4,7,8,9-HpF	< 0.2	< 0.3	< 0.3	< 0.2	< 0.8	< 0.1	< 0.4	< 0.6	< 0.9	< 0.4	< 0.6
non-2,3,7,8-HpF	0.5	1.78	1.18	< 0.3	5.03	0.43	0.46	1.22	1.15	< 0.8	< 1
1,2,3,4,6,7,8-HpD	0.81	1.21	0.44	0.68	23.6	0.82	4.62	10.41	12.6	9.61	4.88
non-2,3,7,8-HpD	< 0.5	0.85	0.31	0.44	9.11	0.37	2.28	5.19	6.97	5.91	2.96
OCDF	< 0.7	1.3	< 0.7	< 0.6	15.3	< 0.6	< 2	< 3	< 3	< 4	< 3
OCDD	5.55	8.85	3.28	5.6	89.1	5.53	15.64	50.0	89.4	74.7	22.6
TE (pg/g)	25.2	0.5	0.4	0.3	18.7	8.50	2.32	11.9	17.1	9.42	9.91
Total TE (PCDDF+PCBs)	200	40.7	24.3	4.24	149.8	101	17.8	131.9	152.7	63.8	140
% TE from PCBs	87.4	98.7	98.2	92.9	87.5	91.6	87.0	91.0	88.8	85.[illegible]	[illegible]